

Ligand-Receptor-based wireless biological nanonetworks

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ABSTRACT

In this work, the Ligand-Receptor model and the way it affects the message exchange process in a biological nanonetwork are under scrutiny. To this aim, a set of efficient simulation schemes for the effects of the number of receptors and the residence time in reception accuracy is presented. The outcome revealed the multifaceted effects of each parameter, rendering different chemotactic release schemes as valuable tools for the enhancement of message delivery between biological nanonodes.

CCS CONCEPTS

• **Networks** → **Network protocols**.

KEYWORDS

biological nanonetworks, ligand-receptor model, chemotaxis, diffusion

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1 INTRODUCTION

Biological nanonetworks constitute the cutting edge in the quest for optimal energy, complexity and biocompatibility since the initial advent of nanotechnology. As traditional communication methods are not directly applicable at the

nanometer scale due to limitations in design and fabrication, new protocols should be tailored to the peculiarities of nanoscale [4]. In this work, a typical communication system consisted of two nodes and one base station (BS) is being analyzed. Nodes communicate by releasing signaling bacteria in the medium and exchanging messages through the changes in concentration under a typical on-off keying method. Bacteria propagate by classical Brownian motion up to the receiver. Upon reaching the vicinity of a receiver and through the ligand-receptor (LR) binding process they are attached to the receiver's periphery and the BS measures the concentration and decodes the message. Although the number of receptors and the period of time each bacterium occupies a receptor site (Residence Time-RT) are of great importance [5], we were motivated by the absence of a relevant study and a set of simulation scenarios were devised in order to shed light in the effects to the reception accuracy [2]. Furthermore, and acting as an additional point of novelty, different input signalling methods were tested under various chemotactic release schemes as a way to bring the system closer to a real life application scenario.

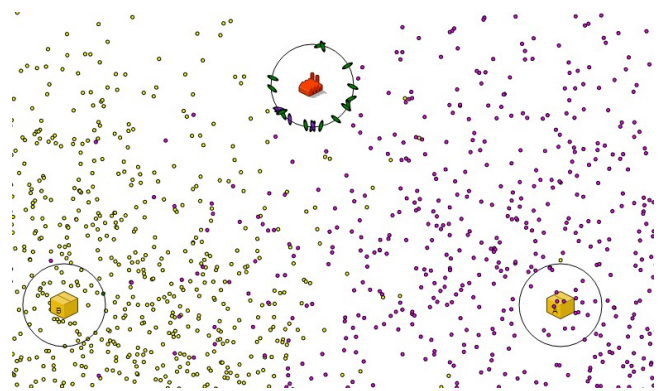


Figure 1: Network topology. Purple and yellow circles are A- and B- type bacteria respectively. On BS's periphery there are bacteria occupying the binding sites.

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2 LIGAND-RECEPTOR MODEL

2.1 Simulation scenarios

The communication system under investigation consisted of one BS placed initially on the perpendicular bisector of the line that connects the two nodes as shown in Fig.1. A base scenario was designed to establish the levels of reception accuracy for the systemic function that was used to discern an incoming pulse from the noise of the channel. Parameters affecting bacterial behaviour were based on [3].

Multiple simulations were conducted to realize the effects of the parameters of the LR model and to validate the effectiveness of different chemotactic release plans. Parameters can be found in [1]. In the base scenario without chemotaxis, each node releases its own type of bacteria every 210s and all entities have 60 receptors in their sensing periphery while the RT was set to 10s. Furthermore, chemoattractant and chemorepellent molecules are released from the BS based on 3 configurations (positive, negative, and alternating between them), as a way to enhance message delivery (bacteria move towards higher concentration areas in case of the former type of molecules or towards lower concentration areas in case of the latter). The Normal distribution with a mean of 150 and standard deviation of 80 was used to generate messages from the nodes in the majority of the scenarios.

2.2 Simulation results

As shown in Fig.2-4 the amount of sensed bacteria for each entity depends heavily on the specific characteristics of each scenario and the reception pattern reflects these variations. More precisely, Fig.2 reflects the reception versus time for the base scenario, resulting in a 90% and 80% reception accuracy for A and B bacteria at the BS, and 80% and 90% for messages received from node A and B respectively. Recep-

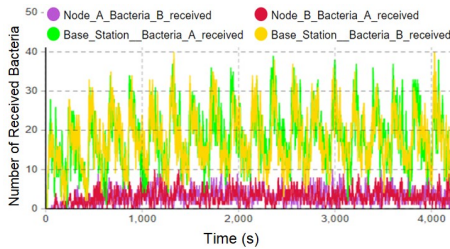


Figure 2: Base scenario, rate 210s, 60 recep, 10s RT

tion accuracy in the case of the Normal distribution with the reduced number of receptors (12) utilizing the alternate chemotaxis scheme is shown in Fig.3. A smaller number for receptors led to a decreased reception accuracy due to the inadequacy of the BS to sense rapid changes in concentration. The alternate policy helped the system to reach an accuracy

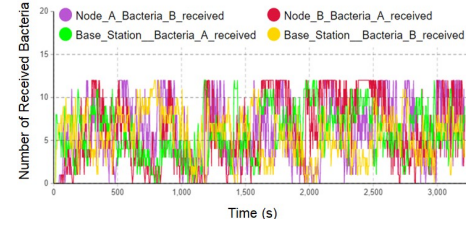


Figure 3: Normal dist(), 12 recep, 10s RT, altrn chemo

of 65% from the 47, 6% of the base scenario. Finally, as shown in Fig.4, reduced RT resulted in less sharper fluctuations of the sensed bacteria with heavier tails, leading to lower message accuracy. Consequently, the BS was able to distinguish the incoming pulses with the use of positive chemotaxis reaching values of 55% from the 46% of the base scenario.

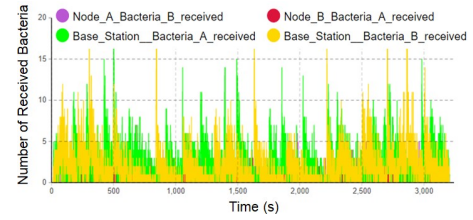


Figure 4: Normal dist(), 60 recep, 2s RT, positive chemo

3 CONCLUSION

This work analysed the effects of the different LR parameters to the reception accuracy of a biological nanonetwork. Simulation results revealed the various implications from these changes in conjunction with different chemotactic schemes. Work in progress encompasses the probing of a self organized biological nanonetwork, able to react directly to the changes of its operational characteristics.

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